

Problem 9

Find all values of k that maximize

$$\binom{50}{k},$$

the number of ways to select k items from 50 distinct items.

Solution

Let $f(k) = \binom{50}{k}$. Using the ratio of consecutive terms:

$$\frac{f(k+1)}{f(k)} = \frac{50-k}{k+1}.$$

The sequence increases while $\frac{50-k}{k+1} > 1 \implies k < 24.5$, and decreases for $k > 25$.

Thus, the maximum occurs at

$$k = 25 \quad (\text{middle index for } n = 50 \text{ even}).$$

Note: The binomial coefficient $\binom{50}{k}$ is largest near the middle because choosing very few items ($k \ll 25$) or all items ($k = 50$) gives far fewer ways. At $k = 50$, the numerator and denominator “cancel out” almost completely, leaving only 1, which is the smallest value. I have a blur idea I may be wrong

Answer

$$\boxed{k = 25}, \quad \binom{50}{25} = \frac{50!}{25!25!}.$$